

The CPU Wars

At the start of the decade, Intel, AMD, Cyrix, NexGen, Motorola, IBM, Texas Instruments, SGS-Thomson, and National Semiconductor were players in the CPU market.

Intel was and continues to be the dominant player. Over the 1990s, Intel developed and released faster and newer versions of its processors at regular intervals. The fifth-generation processor, the Pentium, was out in 1993; the Pentium Pro in 1995; the Pentium MMX and Pentium II were released in 1997; the Pentium III followed in 1999; and the Pentium IV landed up in 2000. Besides these, Intel also made processors for Servers (Xeon), portable PCs (Mobile Pentiums), and Budget PCs (Celerons).

AMD played the underdog to Intel throughout the most of the decade, and was the only real challenger. And, towards the end of the decade, it had even begun to challenge Intel's dominance. AMD's processor releases quickly followed Intel's, and they were usually cheaper than the Intel counterpart. Up to the fifth generation, AMD was only preparing Intel CPU clones, which was cause for litigation with Intel accusing AMD of patent violation with regard to the use of its microcode.

The fifth-generation AMD CPU, labelled the K5, was released in 1995, and was entirely an AMD product. Things truly picked up with the sixth generation: in 1996, AMD merged with chip manufacturer NexGen. The processor NexGen had been working on prior to the takeover had a better design than AMD's own processor. The K6 was based on the NexGen design. The K6 series of processors that came out in 1996 were considered at par with the equivalent Intel processor, besides being cheaper. The K6-II and K6-III upped the ante in the processor war with some of them beating the contemporary Intel product. With the seventh generation Athlon series of processors, which was a result of collaboration with Motorola, AMD surged ahead. The landmark release of the first CPU to breach the 1 GHz barrier was a natural progression.

Cyrix was primarily a chip designer and did not have manufacturing facilities of its own. It initially used the facilities of SGS Thomson and Texas Instruments—and later IBM. Though it did release processors that catered to the low energy consumption niche, especially portable PCs, the significant performance deficiency proved unattractive. Cyrix was bought by National Semiconductor, which later sold it to Via Technologies, a maker of computer chipsets. The Via C3 processor is based on a Cyrix design.

Motorola was also a CPU manufacturer, supplying IBM (PowerPC CPUs), Apple Computers (G4 CPUs) and Commodore (Amiga computers).